SDG BLOCKCHAIN ACCELERATOR

Prototype (PoC) Report

Green Giraffe Digital Traceability & Market Access Pilot

Kurima Programme – Technical Documentation & Test Report

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1. Project Overview

This document presents a **Proof of Concept (PoC)** for a blockchain-based **digital traceability and impact recording system** developed on the **Cardano blockchain**. The prototype demonstrates how agricultural supply-chain actors can **mint, transfer, and verify** traceability tokens that represent batches of farm produce.

The solution is designed to improve **transparency, trust, and market access** for agricultural producers while enabling verifiable impact reporting aligned with the UN Sustainable Development Goals.

Key Features

- **Smart Contracts (Aiken):** Validator scripts enforce authorisation rules, ensuring that only permitted participants can create, update, or transfer traceability records.
- **On-Chain Data Integrity:** Critical traceability data is recorded on-chain, providing immutability, transparency, and auditability.
- **SDG Alignment:**
  - **SDG 2 – Zero Hunger**
  - **SDG 12 – Responsible Consumption and Production**
  - **SDG 13 – Climate Action**
    The system enables verifiable reporting on production practices, supply-chain movement, and environmental impact.

2. Repository Structure

The repository follows standard best practices for Cardano smart contract development:

```
/prototype
```

```
 ├── /contracts   → Aiken validator scripts for traceability logic and
minting policies
 ├── /tests       → Unit tests and emulator-based integration tests
 ├── /scripts     → Cardano CLI scripts for building, signing, and
submitting transactions
 └── /docs        → Technical documentation, deployment notes,
architecture diagrams, and screenshots
```

3. Build Instructions

The smart contracts are implemented using the **Aiken** language.

Prerequisites

- **Aiken CLI:** Version 1.x
- **Cardano Node:** Installed and synchronised (Preprod/Testnet)
- **Cardano CLI:** Available in the system path

Compilation

From the project root directory, run:

```
aiken build
```

Environment variables and keys are not required for compilation, but are necessary when executing deployment scripts.

4. Test Instructions

Testing validates the correctness of datum structures, redeemer logic, and security constraints enforced by the validators.

Execution

```
# Run unit tests
aiken test

# Run emulator-based integration tests
aiken emu
```

Expected Results

- All unit tests pass, confirming correct datum and redeemer validation.
- Emulator tests confirm correct UTxO creation and consumption.
- Invalid redeemers and unauthorised signatures are consistently rejected.

Verified Edge Cases

- Unauthorised token minting attempts
- Invalid or malformed datum structures
- Double-spend prevention through UTxO enforcement

5. Deployment Instructions

Deployment Prerequisites

- Testnet ADA available in the deploying wallet
- Cardano node running and connected to the **Preprod** testnet
- Valid payment and signing keys

Example Deployment Flow

Build Transaction

```
cardano-cli transaction build \
  --testnet-magic 2 \
  --tx-in <UTXO> \
  --tx-out <ADDRESS>+1000000 \
  --change-address <CHANGE_ADDR> \
  --out-file tx.raw
```

Sign Transaction

```
cardano-cli transaction sign \
  --tx-body-file tx.raw \
  --signing-key-file payment.skey \
  --testnet-magic 2 \
  --out-file tx.signed
```

Submit Transaction

```
cardano-cli transaction submit \
  --testnet-magic 2 \
  --tx-file tx.signed
```

The minting policy and spending validators are deployed sequentially as part of the deployment process.

6. Testnet and Emulator Results

The PoC was validated on the **Cardano Preprod testnet**, confirming both functional correctness and execution performance.

Transaction ID	Contract Action	Status	CPU (approx.)	Memory (approx.)	Notes
tx001	Mint Batch Token	Success	320k	910k	Initial batch creation
tx002	Transfer Token	Success	290k	870k	Ownership update validated
tx003	Invalid Redeemer	Failed	—	—	Correctly rejected

7. Dependencies and Environment

- **Aiken CLI:** v1.x
- **Cardano Node:** v8.x
- **Network:** Cardano Preprod Testnet
- **Libraries:** Aiken Standard Library, Cardano CLI
- **Tooling:** Emulator tooling for local testing

8. Demo and Walkthrough

A visual walkthrough demonstrates the core system functionality:

1. Minting a traceability token for a batch of farm produce
2. Transferring the token along the supply chain
3. Verifying on-chain data using a Cardano testnet explorer

Walkthrough Video:
https://www.youtube.com/watch?v=JOa8aQzBHsM&t=23s

9. Remaining Issues and Next Steps

- **Optimisation:** Reduce validator execution units to lower transaction costs
- **Reporting:** Extend off-chain services to support real-time dashboards and analytics
- **Oracles:** Integrate oracle inputs for automated climate and certification data verification
- **Security:** Conduct an independent smart contract security audit
- **Pilot Expansion:** Prepare infrastructure for a scaled pilot using real farmer and aggregator data